Problem

Let m and n be positive integers satisfying

$$\gcd(m,n)+\operatorname{lcm}(m,n)=m+n.$$

Show that one of the numbers is divisible by the other.

Source: **All Russia Mathematics Olympiad 1995**

§1 Converting the Condition into an Algebraic Identity

Let

$$g=\gcd(m,n).$$

Using the standard relation

$$\operatorname{lcm}(m,n)=\frac{mn}{g},$$

the given equation becomes

$$g+\frac{mn}{g}=m+n.$$

Multiplying through by g yields

$$g^2+mn=g(m+n).$$

Rearranging,

$$mn-gm-gn+g^2=0.$$

Factoring,

$$(m-g)(n-g)=0.$$

Therefore,

$$m=g \qquad \text{or} \qquad n=g.$$

§2 Interpreting the Equality

At this stage the algebra is finished, but we must translate the conclusion back into number-theoretic language.

Recall that

$$g=\gcd(m,n).$$

If

$$m=g,$$

then

$$m=\gcd(m,n).$$

Since the greatest common divisor divides each of the two numbers, we have

$$m \mid n.$$

Thus m divides n.

Similarly, if

$$n = g,$$

then

$$n = \gcd(m, n),$$

and therefore

$$n \mid m.$$

Thus n divides m.

§3 The Hidden Meaning of Being the GCD

The key point is that the greatest common divisor is the *largest* positive integer dividing both numbers.

If one of the numbers is itself equal to the gcd, then that number already divides the other. For example,

$$m = \gcd(m, n)$$

means that m divides both m and n, and hence

$$m \mid n.$$

Therefore the two possibilities obtained from

$$(m - g)(n - g) = 0$$

are exactly the two divisibility conclusions required by the problem.

§4 The Final Conclusion

We have shown that

$$(m - g)(n - g) = 0,$$

where $g = \gcd(m, n)$.

Hence either

$$m = g$$

or

$$n = g.$$

In the first case $m \mid n$, and in the second case $n \mid m$.

Therefore one of the two numbers is divisible by the other.

$$\boxed{m \mid n \quad \text{or} \quad n \mid m.}$$